

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | github.com/deepak-glitch |

US-based, F-1 OPT, open to NYC relocation + sponsorship

PROFESSIONAL SUMMARY

Forward-deployed Applied AI engineer (2.5+ yrs) who embeds into client codebases and ships end-to-end agentic systems. Production track record: multi-agent healthcare claims pipeline with schema-validated JSON contracts and audit traces, multi-provider LLM integration shipped to Chrome Web Store, full-stack FastAPI + React/TypeScript multimodal apps, and HIPAA-conscious RAG with ~35% retrieval precision and >30% hallucination reduction. Output velocity: 3 production projects + 5-stage agentic pipeline + ~30% post-deploy defect reduction in 2.5 yrs. Master's degree (Kent State University). Comfortable in AI-native builder cultures (Claude Code, Cursor) and high-velocity client engagements.

CORE COMPETENCIES

Forward Deployed Engineering • Agentic LLM Workflows • Multi-Agent Architecture (Schema Contracts + Audit Traces) • Multi-Provider LLM Integration • RAG + Grounded Retrieval + Eval • Full-Stack Python + TypeScript + React • FastAPI / Flask / Docker • CI/CD + Observability + Structured Logging • HIPAA-Conscious AI Delivery • AI-Assisted Development (Claude / GPT / Cursor)

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - USA - Healthcare AI / Applied AI Consulting

- Embedded into clinical workflows and shipped **agentic LLM workflows** for eligibility, care navigation, and documentation clarification with grounded retrieval, structured outputs, and audit-ready reasoning traces; **~25%** agent response stability gain.
- Built **RAG systems** for healthcare knowledge retrieval and documentation search using recursive semantic chunking and transformer embeddings; **~35%** retrieval precision gain, **>30%** irrelevant retrieval reduction, retrieval-grounded alignment **>90%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement, and prioritization; **15-20%** recall gains on high-risk categories with precision **>90%**.
- Packaged ML/LLM inference as **FastAPI/Flask** services in **Docker** with structured logging and load simulation; **~30%** post-deploy defect reduction.
- Maintained HIPAA-conscious data governance: de-identification, lineage docs, evaluation audit trails, stakeholder-facing system-limitation documentation.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India - Oil & Gas / Enterprise ERP

- Owned **Jenkins CI/CD** for Synthesis Order-to-Cash schema/SP deployments with rollback checkpoints; **>30%** deployment error reduction, **~35-40%** release cycle improvement.
- Tuned T-SQL stored procedures for contracts/nominations/allocations/invoicing; **~20%** query time reduction, **~25%** latency reduction.
- Built SQL Server DMV + Grafana dashboards for long-running queries, deadlocks, CPU/I/O contention; **~25%** incident recurrence cut, **~30%** faster RCA.
- Supported RBAC + audit logging for financial modules in compliance-sensitive oil & gas environment.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Hyderabad, India - Nonprofit Tech / Video Analytics / Applied NLP

- Shipped **transformer-based video summarization** across 5,000+ sessions with hierarchical summarization and timestamp-aligned clip extraction; **60-70%** review time reduction, **~85%** alignment with human-curated highlights.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); **~30%** hallucination reduction, **>85%** contextual accuracy in controlled tests.
- Deployed as **Flask API** with lightweight web UI for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization → Validation → Consistency → Duplicate Detection → Fraud Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity search for duplicate claims; explainable risk scoring with audit-ready reasoning traces.

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation Shipped

Full-stack browser extension built solo - Manifest V3, content scripts, service workers, captureVisibleTab. Integrates 4 AI vision providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-section capture for tall webtoon panels with coordinate remapping, 7-day translation cache, per-domain selector configs for 29 sites, narrowed host permissions. Provider-aware payload normalization (WebP cloud / JPEG Ollama).

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind interpreting dreams and synthesizing dreamscapes via coordinated multimodal APIs (Perplexity Sonar interpretation + Replicate visuals). Structured prompt transformation layers improved contextual alignment ~30% and first-pass image success ~25-30% over naive direct-prompt orchestration.

Driver Drowsiness Detection System - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN pipeline with unified YOLOv8 detection-and-classification; ~30% inference latency reduction. Sliding-window confidence aggregation suppressed blink-driven false positives ~25%; adaptive frame skipping and NMS tuning for stable real-time operation via OpenCV.

Agentic Pixel Character Synthesis & Animation Engine

Phased generative AI system for identity-consistent pixel character synthesis with sprite-sheet export. Stable Diffusion fine-tuning + LoRA, ControlNet + OpenPose/MediaPipe pose conditioning, latent-space animation smoothness, and an LLM-based agent orchestrator that decomposes high-level prompts into generation tasks. PyTorch + Diffusers + FastAPI + Docker.

E-Farming Digital Marketplace Founder

Solo-shipped PHP/MySQL/Bootstrap AgriTech marketplace connecting small-scale farmers with buyers; onboarded 80-120 active users in phase one. End-to-end ownership: schema design, application logic, UX, deployment.

EDUCATION

Master's Degree - Kent State University, Ohio, USA Aug 2023 - May 2025

CERTIFICATIONS

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, SQL (T-SQL, PostgreSQL, SQLite), C++ (Arduino), PHP

LLM & Agentic: LLM application development, multi-agent architecture, RAG, structured outputs, prompt engineering, eval pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search

ML & CV: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex, YOLOv8, OpenCV, ControlNet, Stable Diffusion

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, NetworkX

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability, structured logging, AI-assisted dev (Claude Code / Cursor)

Domains: Forward Deployed Engineering, Applied AI, Healthcare AI (HIPAA-conscious), Enterprise AI, Workflow Automation, Computer Vision, Multimodal Systems